

The background of the entire page is decorated with large, overlapping, wavy shapes in various shades of teal and green. These shapes create a sense of movement and depth, framing the central text.

RETAIL SALES DATA SYSTEM

SYSTEM REQUIREMENTS & SPECIFICATION

Contents

1. System Description.....	3
1.1 Introduction	3
1.2 Product Aim	3
1.3 Definitions	3
1.4 Contribution Table	4
2. System Requirements	4
2.1 System Description.....	4
2.2 Informational Requirements	4
2.3 Functional Requirements	4
2.4 Non-Functional Requirements	5
3. Data Warehouse Design	5
3.1 Context Diagram.....	5
3.2 Entity Relationship Design	6
3.3 Data Dictionary	8
4. Conclusion	11
5. Appendix.....	11
Appendix A: System Context Diagram	11
Appendix B: Entity Relationship Design	12

1. System Description

1.1 Introduction

Manual monitoring of sales data in retail stores is time-consuming and more prone to errors. This process makes it difficult for managers and businesses to accurately measure store key performance indicators and identify trending products. This project introduces a Retail Sales Data System that automates logging successful transactions allowing for more efficient business and KPI monitoring.

The report presents the Retail Sales Data System, highlighting system purpose and requirements to address current challenges. Additionally, it proposes a high-level context diagram and a data warehouse design, illustrating the interaction between external entities and data storage.

1.2 Product Aim

The aim of Retail Sales Data System is to allow businesses make timely decisions and monitor performance more efficiently. The system aims to:

- Automate the injection of sales from various store locations through bulk csv uploads
- Provide near real-time insights into sales performance for informed and efficient business decisions
- Enable tracking of KPI metrics such as revenue, product performance and location performance
- Display clear data visualisation and reports through a user-friendly dashboard
- Introduce centralised data management by combining all store data into one unified system

1.3 Definitions

Acronym	Definition
CSV	Comma-Separated Values
ETL	Extract, Transform, Load
GUI	Graphical User-Interface
KPI	Key Performance Indicator
SQL	Structured Query Language

Table 1.1: Definitions

1.4 Contribution Table

Group Member Name	Student ID	Document Sections Contributed To
Joshua Tangendjaja	24442143	Definitions, Informational Requirements, Functional Requirements, Non-Functional Requirements, Entity-Relationship Diagram, Reviewing, Editing
Michaela Sabio	24632123	Introduction, Product Aim, Context Diagram and Description, Reviewing, Editing
Vu Tan Huy Nguyen	14517484	Conclusion, Data dictionary, Reviewing, Editing

Table1.2: Contribution table

2. System Requirements

2.1 System Description

Retail businesses generate large amounts of sales data from multiple locations, but this data is often difficult to manage and analyse effectively. Our system provides tools for monitoring sales performance and generating insights from collected data. Its primary purpose is to support executives and company managers in understanding sales trends and making informed business decisions.

2.2 Informational Requirements

Informational Requirements ID	Description
IR 2.2.01	The system users will be higher level executives and authorised staff
IR 2.2.02	The system should use local files for the data lake
IR 2.2.03	The system must complete the ETL process using Python Pandas
IR 2.2.04	The system should use Microsoft SQL Server for the data warehouse
IR 2.2.05	Data extraction will be executed using SQL
IR 2.2.06	The system should accept sales data in CSV file formats submitted by the individual stores

Table2.1: Information requirement table

2.3 Functional Requirements

Functional Requirements ID	Description	Priority
----------------------------	-------------	----------

FR 2.3.01	The system should allow users to upload CSV files containing sales data.	Mandatory
FR 2.3.02	The system should store uploaded sales data into structured database tables	Mandatory
FR 2.3.03	The system should allow users to view sales performance by category (i.e. item or location)	Mandatory
FR 2.3.04	The system must allow users to generate product performance reports based on sales volume and revenue	Mandatory
FR 2.3.05	The system should generate summaries and reports of sales trends through a GUI	Recommended

Table2.2: Functional requirements table

2.4 Non-Functional Requirements

Non-Functional Requirements ID	Description	Purpose	Priority
NFR 2.4.01	Access to sales data should be restricted to authorised users	Security	Mandatory
NFR 2.4.02	The system should be designed to support future expansion in the number of stores and data volume	Scalability	Recommended
NFR 2.4.03	The system shouldn't load for significant time when processing large sales datasets	Usability	Recommended
NFR 2.4.04	System maintenance should be implemented without disruption	Maintenance	Recommended
NFR 2.4.05	The system should be uncomplicated, allowing users to utilize all the features without difficulty	Usability	Recommended

Table2.3: Non-Functional requirements table

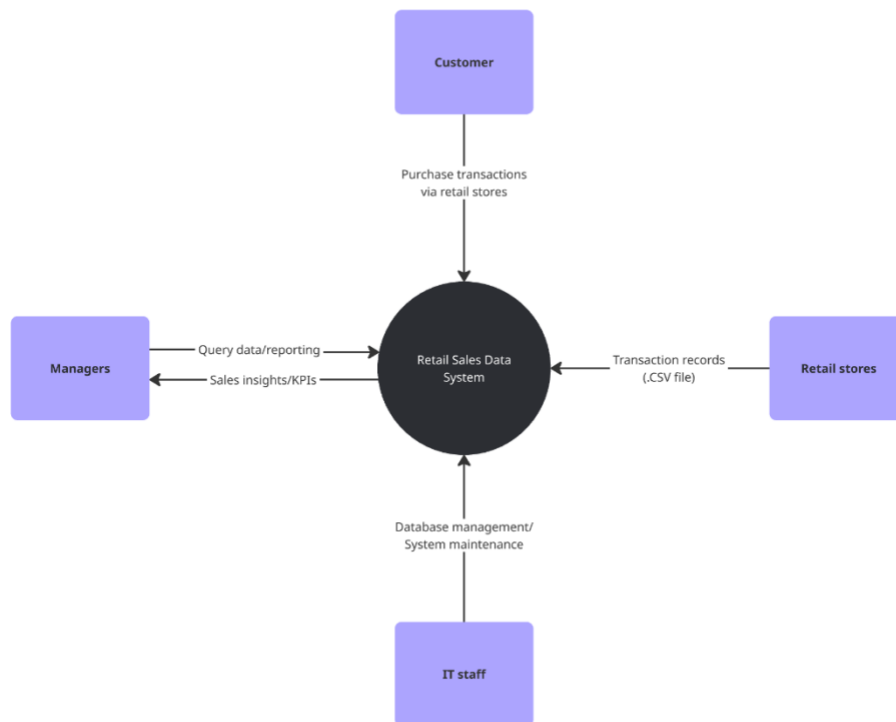
3. Data Warehouse Design

3.1 Context Diagram

The context diagram for the Retail Sales Data System addresses the main interaction between the central system and its external entities. The customer initiates the process by purchasing transactions through retail stores' point of sales system. These transaction records are subsequently sent into the central system as CSV files, which serve as the main data source. IT staff communicate with the system to perform

database administration and system maintenance. Store managers engage with the system through data queries and report request, in which enables the system to provide them with vital sales insights and KPIs to help them make well-informed business decisions.

Figure 1. System Context Diagram.



3.2 Entity Relationship Design

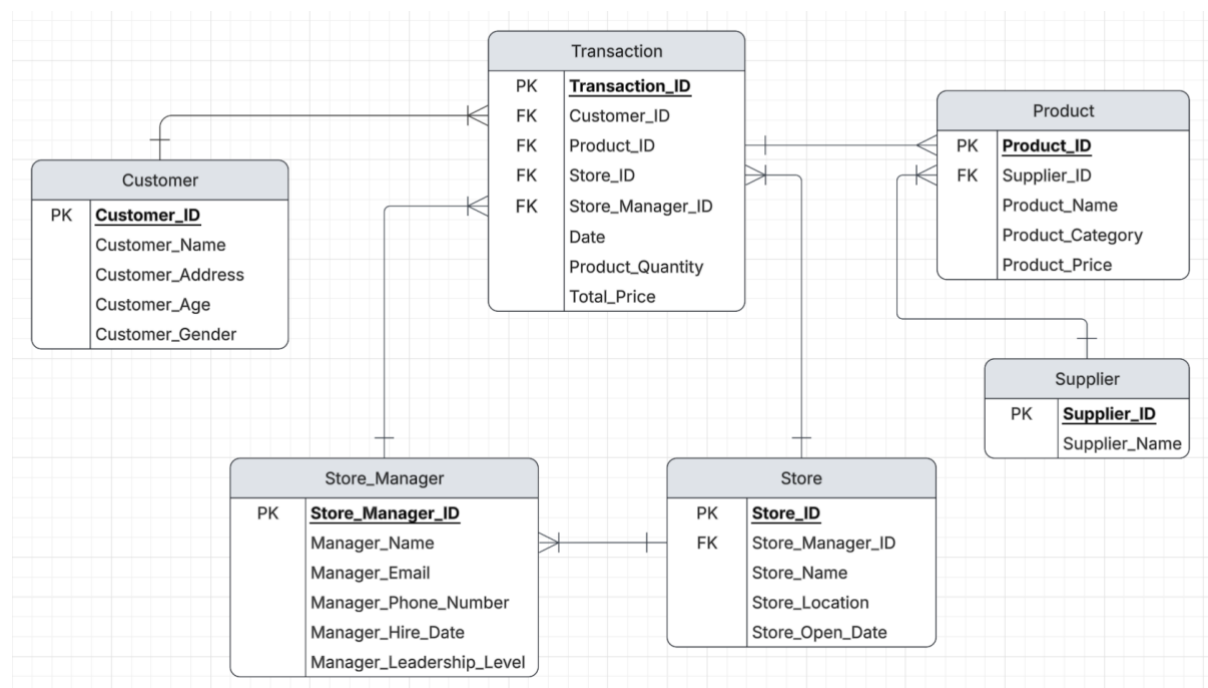


Figure 2. Entity Relationship Design

Entity	Primary Key (PK)	Foreign Key (FK)
Transaction	Transaction_ID	Customer_ID, Product_ID, Store_ID, Store_Manager_ID
Product	Product_ID	Supplier_ID
Store	Store_ID	Store_Manager_ID
Store_Manager	Store_Manager_ID	N/A
Customer	Customer_ID	N/A
Supplier	Supplier_ID	NA

Table3.1: Entity key relationship table

3.3 Data Dictionary

Entity: Transaction

Attribute	Data type	Description	Example
<u>Transaction_ID</u>	Char	A unique 10 letters and number to identify a transaction	Tr43767583
Customer_ID	Int	A unique 8 numbers to identify customers who made the transaction	82324232
Product_ID	Int	A unique 11 numbers to identify a product	45367278965
Store_ID	Char	A unique 4 letters and numbers to identify a store	SY01
Store_Manager_ID	Char	A unique 8 letter and numbers to identify manager ID in which first 4 character is Store ID	SY019990
Date	Datetime	When the transaction is made	2026-03-29
Product_Quantity	Int	Number of products in one transaction	3
Total_Price	Float	Total price in one transaction	50.79

Table3.2: Transaction dictionary

Entity: Store

Attribute	Data type	Description	Example
<u>Store_ID</u>	Char	A unique 8 letters and numbers to identify manager ID in which first 4 character is Store ID	SY019990
<u>Store manager ID</u>	Char	Short text that displays manager's name	John

Store_Name	Char	Short text that displays store's name	Harbour City Flagship
Store_Location	Char	Short text that displays store's address	15 Broadway, Ultimo NSW 2007
Store_Open_Date	Datetime	A date showing when the store was opened	2025-08-15

Table3.3: Store dictionary

Entity: Customer

Attribute	Data type	Description	Example
Customer_ID	Int	A unique 8 numbers to identify customers who made the transaction.	82324232
Customer_Name	Char	Full name of the customer	Jane Doe
Customer_Address	Char	A short text that displays customer's address	123 Main St, Sydney
Customer_Age	Int	A digit that display customer's age	20
Customer_Gender	Char	Display customer's gender	Female

Table3.4: Customer dictionary

Entity: Store Manager

Attribute	Data type	Description	Example
Store_Manager_ID	Char	A unique 8 letters and numbers to identify manager ID in which first 4 character is Store ID	SY019990
Manager_Name	Char	Full name of the manager	John Smith
Manager_Email	Char	The email address for the manager	j.smith@retail.com

Manager_Phone_Number	Int	The 10-digit mobile or office contact number.	0412345678
Manager_Hire_Date	Datetime	The official date that the manager joined the company.	2026-05-08
Manager_Leadership_Level	Char	Text identifying the manager's title	Senior Manager

Table3.5: Store manager dictionary

Entity: Supplier

Attribute	Data type	Description	Example
<u>Supplier_ID</u>	Char	A unique character to identify a supplier	SUP99
Supplier_Name	Char	A short text that represents supplier's trademark	Australia Logistics Ltd

Table3.6: Supplier dictionary

Entity: Product

Attribute	Data type	Description	Example
<u>Product_ID</u>	Int	A unique 11 numbers to identify a product	10000000055
Supplier_ID	Char	A unique character to identify a supplier	SUP-99
Product_Name	Char	To identify the product	Pepsi
Product_Category	Char	An attribute that is used to group similar products together	Dairy
Product_Price	Float	The price of a single product	90.32

Table3.7: Product dictionary

4. Conclusion

Through offering a centralized, automated structure for data administration and KPI tracking, the Retail Sales Data System successfully address the inefficiencies of manual sales tracking system. The system converts fragmented store data into meaningful, real-time information for executive decision-making by combining a raw data lake with an organized Microsoft SQL server data warehouse and utilizing Python Pandas for reliable ETL processing. In the end, this integrated strategy not only improves operational accuracy and scalability but also gives companies the capacity to recognize market trends and maximize site performance through an intuitive, data-driven interface.

In conclusion, the Retail Sales Data System is a strategic asset rather than simply a tracking tool. The technology enables retail managers and executives to move from reactive maintenance to proactive, data-driven decision-making by combining a raw data lake with an organized SQL based warehouse and an intuitive graphical user interface. This system's development will greatly operational accuracy and understanding of market trends

5. Appendix

Appendix A: System Context Diagram

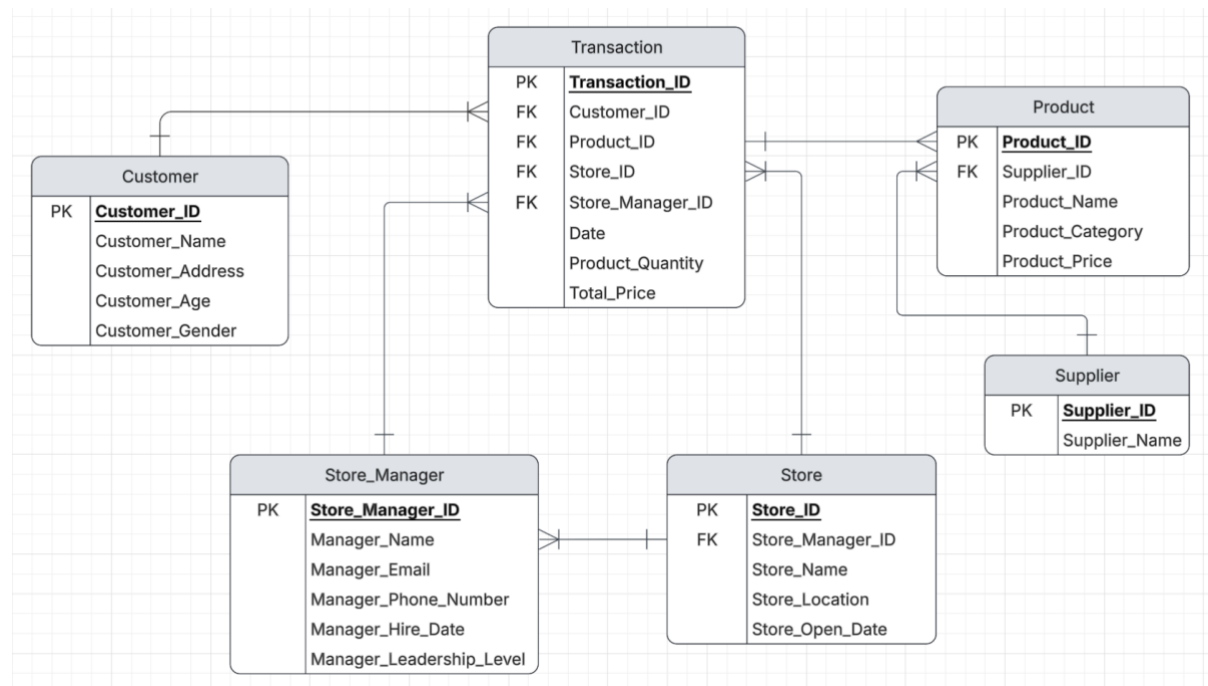


Digital diagram:

<https://miro.com/welcomeonboard/UjNpbCtxWUtBbGNGZ1lKNWVPY1lMcDFUZI0dElTTFNSdTZWR2poSUZxUVZzWVpUbmh6eXZpUk1SQm56NENHVzVzZEI3SThHbjBsSjVKQ>

m50OU1HSnBkUVgzbmJOaG94L0hDL0hOMnpRdGZKTnFrZWdGem0yRFN1WklHMkhrS
UNnbHpza3F6REdEcmNpNEFOMmJXWXBPT0hdjE=?share_link_id=96320938168

Appendix B: Entity Relationship Design



Digital diagram: https://lucid.app/lucidchart/afe6914d-4053-4dbb-a260-b222875a6b4f/edit?viewport_loc=-1198%2C-353%2C1141%2C1388%2C0_0&invitationId=inv_e828e90d-513b-4316-9be8-d5bb3c233668